

Question 1. (7 marks)

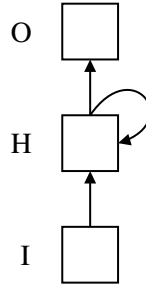
Consider the following input array:

$$\mathbf{x} = (1, 0, 2, -3, 2, -1, -4, 5, 7, 9).$$

- (a) (2 marks) Apply the kernel $\mathbf{w} = (\frac{1}{2}, 1, -\frac{1}{2})$ to $\mathbf{x}$ using a bias of 1 and *relu* activation function, and then apply max-pooling using a pooling window of size 2 and with a stride of 2.
- (b) (3 marks) What are the advantages of using a convolutional layer in a neural network?
- (c) (2 marks) If a kernel $\mathbf{w} = (w_1, \dots, w_k)$ is applied to an input vector $\mathbf{x} = (x_1, \dots, x_m)$ using stride s , and no padding is used, give the length of the output vector.

Question 2. (4 marks)

Consider a recurrent neural network with structure as shown in the diagram below, that has 3 input nodes, 3 nodes in the hidden layer and one output node.



Let the weight matrices and biases be as follows:

$$W_X = \begin{bmatrix} -1 & 2 & -3 \\ -2 & -4 & 2 \\ 3 & 0 & -1 \end{bmatrix}, \quad W_H = \begin{bmatrix} 2 & 0 & -3 \\ -1 & 2 & 1 \\ 0 & -4 & 3 \end{bmatrix}, \quad W_O = \begin{bmatrix} 0.3 \\ -0.2 \\ 0.1 \end{bmatrix}$$
$$\mathbf{b}_H = (3, 1, 2), \quad \mathbf{b}_O = (0.5).$$

The hidden layer uses *relu* activation function and the output layer uses *sigmoid* activation function.

Compute the outputs of the network if the sequence $\mathbf{x}$ is input, with $\mathbf{x}(1) = (2, 1, 1)$, $\mathbf{x}(2) = (-1, 1, 2)$.

Question 3. (4 marks)

Give pseudocode for the *k*-MEANS ALGORITHM, as applied to a dataset S of points in $\mathbb{R}^m$.

Question 4. (5 marks)

- (a) (4 marks) Use Hierarchical clustering with the **centroid linkage** on the following dataset.
- (b) (1 mark) Draw the dendrogram for the clustering obtained.

	x_1	x_2
A	-2	-2
B	-1	-1
C	0	4
D	1	0
E	2	2
F	2	3

Question 5. (4 marks)

- (a) (2 marks) With the aid of a diagram, describe the structure of an undercomplete autoencoder.
- (b) (1 marks) Give the loss function that would typically be used to train an autoencoder.
- (c) (1 marks) Give the loss function that would be used to train a denoising autoencoder.

Question 6. (6 marks)

- (a) (2 marks) Give a diagram illustrating the structure of a generative adversarial network.
- (b) A generative adversarial network is to be trained using the following minimax loss function:

$$\mathbb{E}_{\mathbf{x}}[\log(D(\mathbf{x}))] + \mathbb{E}_{\mathbf{z}}[\log(1 - D(G(\mathbf{z})))].$$

- (i) (2 marks) Describe the process of discriminator training.
- (ii) (2 marks) Describe the process of generator training.

Question 7. (4 marks)

Give pseudocode for the algorithm SARSA, as applied to an MDP with set S of states and set A of actions. Include a description of the ϵ -greedy policy in your answer.

Question 8. (6 marks)

Consider the MDP illustrated in the diagram below.

s_0	s_1	s_2	s_3	s_4
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The set of states is $S = \{s_0, s_1, s_2, s_3, s_4\}$ and the set of actions available at each state is $A = \{\ell, r\}$. The action ℓ moves the agent one state to the left with probability 0.2 and leaves the agent in the current state with probability 0.8. The action r moves the agent one state to the right with probability 0.6 and leaves the agent in the current state with probability 0.4. The states s_0 and s_4 are both goal states and any action that results in getting to either goal state gives a reward of 10 and ends the episode. An action that does not move the agent into a goal state gives a reward of -1 .

For example, $P(s_1, \ell, s_0) = 0.2$ and $P(s_1, \ell, s_1) = 0.8$, and $R(s_1, \ell, s_0) = 10$ and $R(s_1, \ell, s_1) = -1$.

The VALUE ITERATION ALGORITHM is being used to find the optimal policy for the above MDP.

After 2 iterations, the values for V are:

0	2.16	1.96	7.84	0
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(a) (4 marks) Perform one more iteration of the VALUE ITERATION ALGORITHM using $\gamma = 1$.

Show all your working.

(b) (2 marks) Calculate the action to take in state s_1 determined by the V table obtained in (a).

Show all your working.